



College of Computing and Informatics
Department of Computer Science

Course Syllabus

Course Title **Discrete Structures**

Course Code
1501279

Spring Semester, 2022 / 2023

Course Syllabus

Course Title	Discrete Structures		
Course Code	1501279		
Credit Hours	3		
Pre-requisite(s)	1440131 Calculus I		
Co-requisite(s)	None		
Semester	Spring	Year	2022/2023
Instructors Name	Prof. Sebti FOUFOU		
Office Location	M5-212		
Tel. No.			
Email	sfoufou@sharjah.ac.ae		
Lecture Times	Monday and Wednesday 11:00 – 12:30		
Office Hours	Tuesday & Thursday : 11:30 – 11:30		

Course Description (as in the catalogue):

This course emphasizes the representations of numbers, arithmetic modulo, radix r representation of integers, change of radix. Negative and rational numbers. Sets, one-to-one correspondence, properties of union, intersection, and complement, countable and uncountable sets. Functions: Injective, surjective, and bijective functions. Mathematical Induction, proof by contradiction. Combinatorics: Multiplication rule, pigeonhole principle, Recurrence relations. Fundamentals of logic, truth tables, conjunction, disjunction, and negation, Boolean functions and disjunctive normal form. Logic circuits. Graphs Theory: Introduction, Paths and connectedness, Eulerian and Hamiltonian Graphs, Graph Isomorphisms. Trees: Spanning trees, Binary Search Trees, Huffman Code.

Course Objectives/Goals:

The goals of this course are to:

1. Introduce basic principles and techniques of discrete structures and their algorithms for computer Sciences applications.
2. Develop the ability to manipulate abstract structures such as sets, graphs, trees, relations, and functions.
3. Understand and implement different proof techniques and mathematical reasoning,

including formal logic.

4. Introduce principles of graphs and trees and their applications to various scientific fields.

Course Learning Outcomes:

By the end of successful completion of this course, the student will be able to:

- A. Apply knowledge of mathematics of discrete structures in modeling applications.
- B. Apply the standard propositional logic connectives and be able to convert logical expressions into conjunctive and disjunctive normal forms and use the truth tables.
- C. Use proof by contradiction and proof by induction techniques.
- D. Identify applications that require the use of sets and functions.
- E. Apply different counting techniques.
- F. Solve relevant problems using graph theory.

Alignment of Course Student Learning Outcomes to the BSCS Program Student Learning Outcomes

Computer Science Program SLOs		Course SLOs
Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	Analyze	- Use proof by contradiction and proof by induction techniques. - Identify applications that require the use of sets and functions.
	Apply	- Apply knowledge of mathematics of discrete structures in modeling applications. - Apply the standard propositional logic connectives and be able to convert logical expressions into conjunctive and disjunctive normal forms and use the truth tables. - Apply different counting techniques.
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	Design	-
	Implement	- Solve relevant problems using graph theory.
	Evaluate	-
Communicate effectively in a variety of professional contexts.		-
Describe professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.		-
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.		-
Apply computer science theory and software development fundamentals to produce computing-based solutions.	Theory	-
	Development	-

Alignment of Course Student Learning Outcomes to the BSBI Program Student Learning Outcomes

PLOs	CLOs
Apply knowledge of mathematics, computer science, computer engineering, and computational biology appropriate to the discipline.	- Apply knowledge of mathematics of discrete structures in modeling applications. - Apply the standard propositional logic connectives and be able to convert logical expressions into conjunctive and disjunctive normal forms and use the truth tables. - Apply different counting techniques. - Solve relevant problems using graph theory.
Analyze a problem, identify and define the computing requirements appropriate to its solution.	- Use proof by contradiction and proof by induction techniques. - Identify applications that require the use of sets and functions.
Design and evaluate a software system, process, or program to meet desired needs.	-
Function effectively on teams and communicate effectively with a range of audiences.	-
Identify professional, ethical, legal, security and social issues and responsibilities related to biomedical informatics.	-
Analyze the local and global impact of biomedical informatics on individuals, organizations, and society.	-
Apply design and development principles in the construction of software and informatics- based solutions of varying complexity.	-

Weekly Distribution of Course Topics/Contents

Week	Topic	Comments*	Course SLO
1 - 2	Representation of numbers, modulo arithmetic, Radix representation of integers, change of radix. Negative and rational numbers. Quiz #1	Introduce the students to different number systems.	A
3.	Boolean functions and logic circuits. Fundamentals of Logic: Statements, connectives, and truth tables. How can English sentences be translated into a logical expression?	Introduce the students to the concept of logic.	B
4.	Logical equivalence, conjunction, disjunction, and negation. Tautology and Contradiction Boolean functions and disjunctive normal form. Quiz #2 Due date of HW #1		C

5	Mathematical Proofs: direct proof, proof by contrapositive, Principle of mathematical induction. Proof by contradiction.	Introduce the students to different methods of proof.	C
6.	Sets, one-to-one correspondence, properties of union, intersection, and complement, countable and uncountable sets. Due date of HW #2	Introduce the students to sets, relations, and functions.	D
7.	Relations: Their Properties, Representing Relations, and Equivalence relations. Recurrence relations. Functions: Injective, surjective, and bijective functions. Growth of functions Quiz #3		
8.	Combinatorics: Multiplication rule, pigeonhole principle.	Introduce the students to the concepts of combinatorics, and number theory.	E
9.	Numbers theory (divisibility, greatest common divisor, Euclidean Algorithm, modular arithmetic and congruence Quiz #4 Due date of HW #3		
10.	Graphs Theory: Definitions, Graph models, directed graphs, undirected graphs,	Introduce the students to the graph theory.	F
11.	Representing graphs, connected graphs, disconnected graphs, Paths and connectedness, Eulerian and Hamiltonian Graphs. Quiz #5		
12.	Graph Isomorphisms, counting paths, shortest paths, bipartite graphs		
13.	Trees: properties of trees, Spanning trees, Minimum spanning trees. Due date of HW #4		
14.	Trees: Binary Search Trees, Huffman Codes. Quiz #6		
15.	Revision.		
16.	Final Exam		

Information on out-of-class assignments with due dates for submission

Assignment/Activity	Due Date	Comments
HW #1	Weeks: 1 - 4	Covers number systems and logic.
HW #2	Weeks: 5 - 6	Covers logic, proofs and sets.
HW #3	Weeks: 7 - 9	Covers relations, functions, combinatorics, and number theory.
HW #4	Weeks: 10 - 14	Covers graph theory and trees.

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)	Date	Weight (%)
Assignments	Weeks: 4, 6, 9, 13	15
Midterm	Week 8	25
Quizzes	Weeks: 1, 4, 7, 9, 11, 14	15
Final Exam	Week 16	45
Total		100

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators
Demonstrate a working knowledge of mathematics of discrete structures, and the application of discrete mathematics to modeling in computer science.	- Live Online Lectures Recordings - Problem solving	- Quizzes - Exams - Assignments	Course assessment
Apply the standard propositional logic connectives and be able to convert logical expressions into conjunctive and disjunctive normal form, and use the truth tables.	- Live Online Lectures Recordings - Problem solving	- Quizzes - Exams - Assignments	Course assessment
Demonstrate competence in several styles of proof, including proof by contradiction and proof by induction.	- Live Online Lectures Recordings - Problem solving	- Quizzes - Exams - Assignments	Course assessment
Demonstrate competence in using and working with sets and functions.	- Live Online Lectures Recordings - Problem solving	- Quizzes - Exams - Assignments	Course assessment
Apply different counting techniques.	- Live Online Lectures Recordings - Problem solving	- Quizzes - Exams - Assignments	Course assessment

Apply the graph theory to solve practical problems.	- Live Online Lectures Recordings - Problem solving	- Quizzes - Exams - Assignments	Course assessment
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Teaching and Learning Resources:

Text Book(s):

Kenneth H. Rosen, *Discrete Mathematics and Its Applications*, 8th edition, McGraw-Hill, 2007.

Library Reference: QA39.2.R6542005.

Other Resources:

James A. Anderson, *Discrete Mathematics*, Prentice Hall, 2001. Library Reference: QA39.2.A534 2001.

John Truss, *Discrete Mathematics for Computer Scientists*, 2nd edition, Addison Wesley, 1999.

Library Reference: QA76.9.M35.T78 1999.

Steven Roman, *An Introduction to Discrete Mathematics*, 2nd edition, HBJ, 1989. (Required).

Library Reference: QA76.9.M35.R66 1989.

Attendance policy:

Attendance is compulsory. A student missing 10% (6 hours) of the total allocated course hours will receive 1st warning notice and a student missing 15% (9 hours) will receive 2nd warning notice. A student missing 20% (12 hours or more) will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Assessment Methods:

The course includes two lectures per week for 15 weeks, 6 theory quizzes, and 4 assignments. The quizzes and assignments are on the same level of difficulty with the midterm and final exams. Different sections are given different problems to solve and solution of all problems are posted to all sections after the end of each quiz. The assignments were designed to enforce the problem solving and critical thinking skills.

Academic Integrity, Collaboration

All students are expected to maintain the utmost level of academic integrity. Any attempts to cheat will lead to appropriate sanctions. Assignments should be solved mostly individually. Substantial collaboration in writing up solutions is not appropriate. Nor is of course copying

classmates' solutions. It is also never appropriate to search for solutions online and try to pass them off as your own work.

You may discuss problems with your fellow students and the course staff, but you must write your own solutions. You may also consult any publications. Clearly cite any collaborators or outside sources. You may not give or receive help from anyone or any publications.

Digital Skills Mapping:

The following table shows how digital skills are developed during the course by covering the given topics and measuring the skill using different assessment techniques.

Digital Skill	Topics Covered	Assessment Tool(s)
1. Critical thinking and analysis	Boolean functions, and Logical Equivalence	- Assignments - Quizzes - Exams
2. Reasoning, problem-solving, and ideation	Representation of numbers, Graphs Theory, Graph Isomorphism, and Trees	- Assignments - Quizzes - Exams